

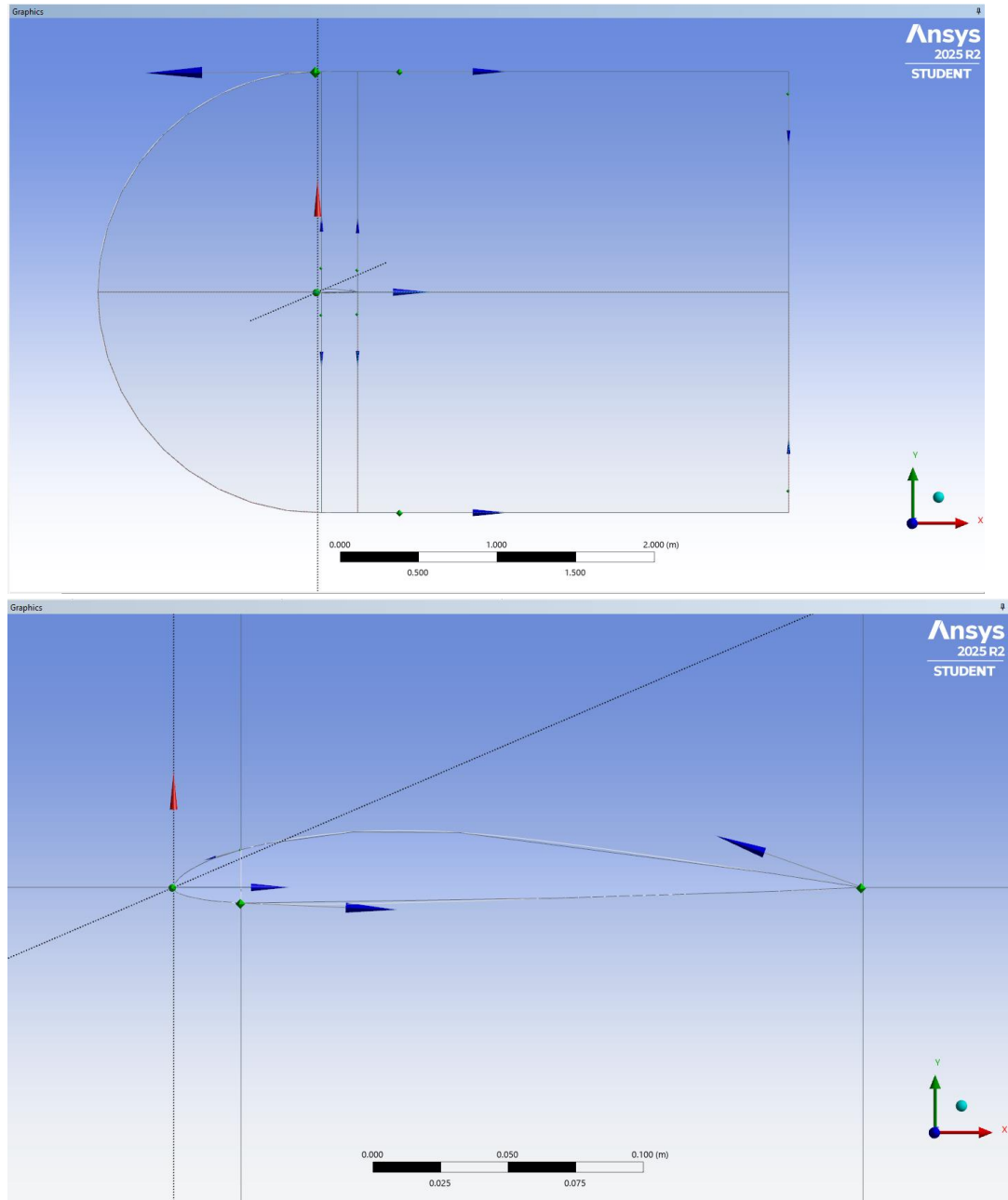
CFD FINAL PROJECT

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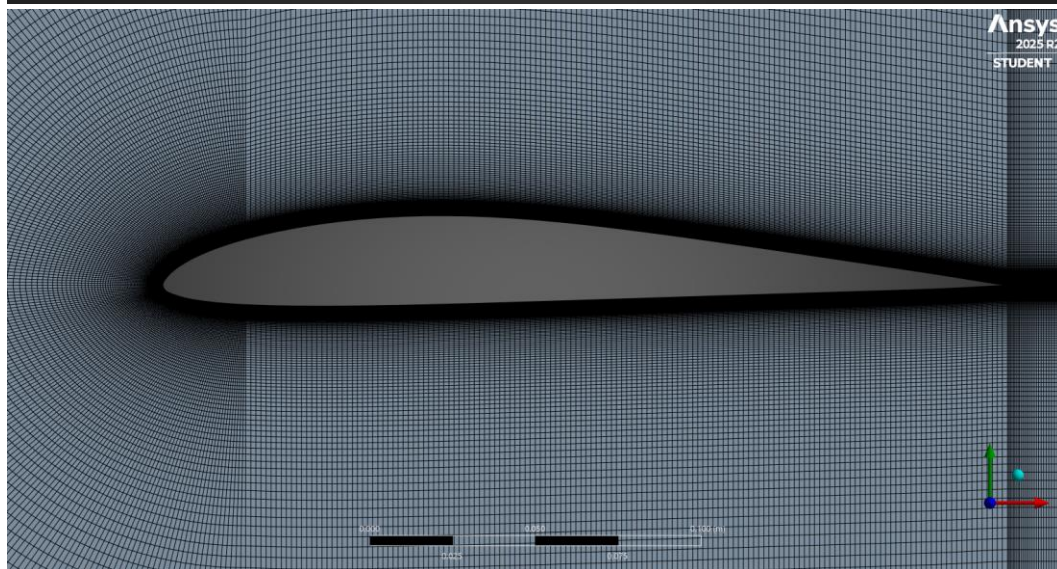
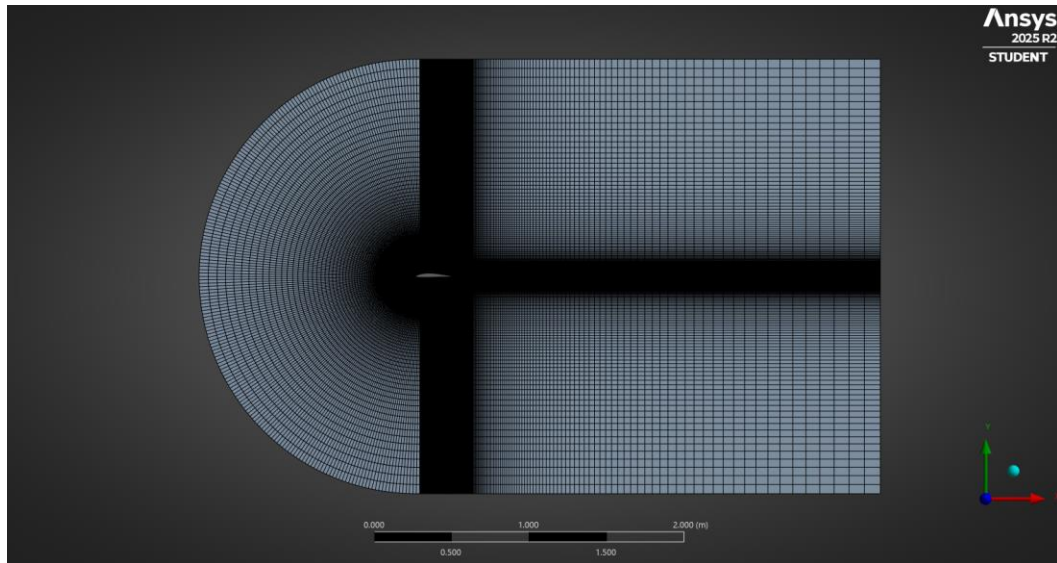
1. Mesh Generation Airfoil

Airfoil Type: Epler205

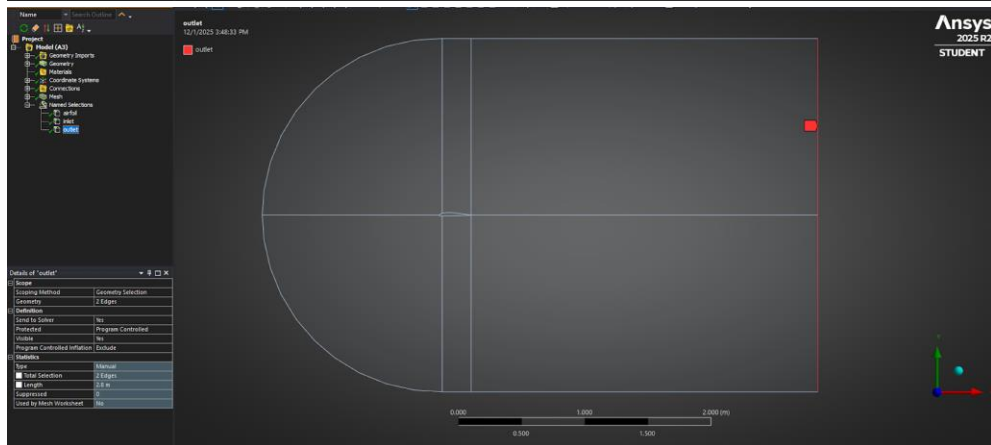
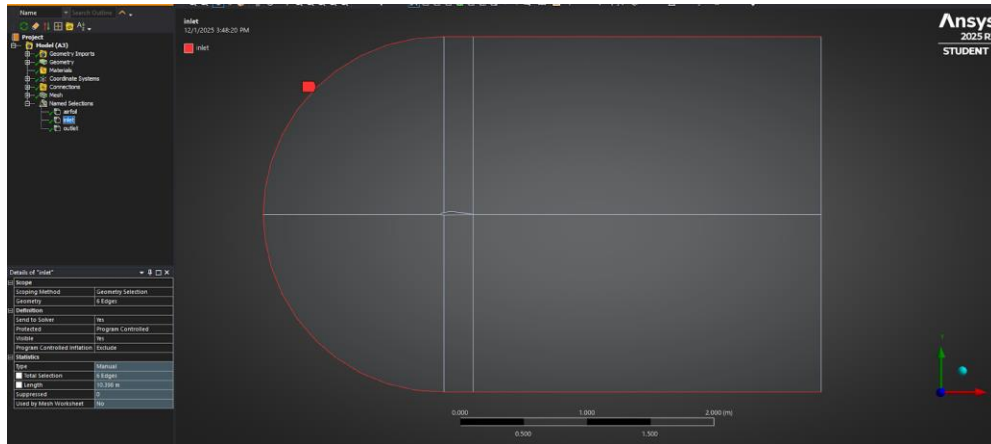
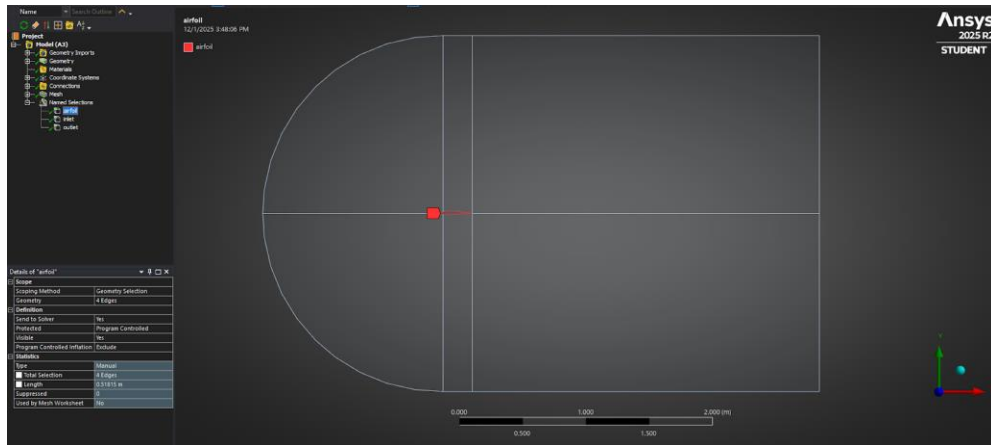
1.1) Airfoil Geometry



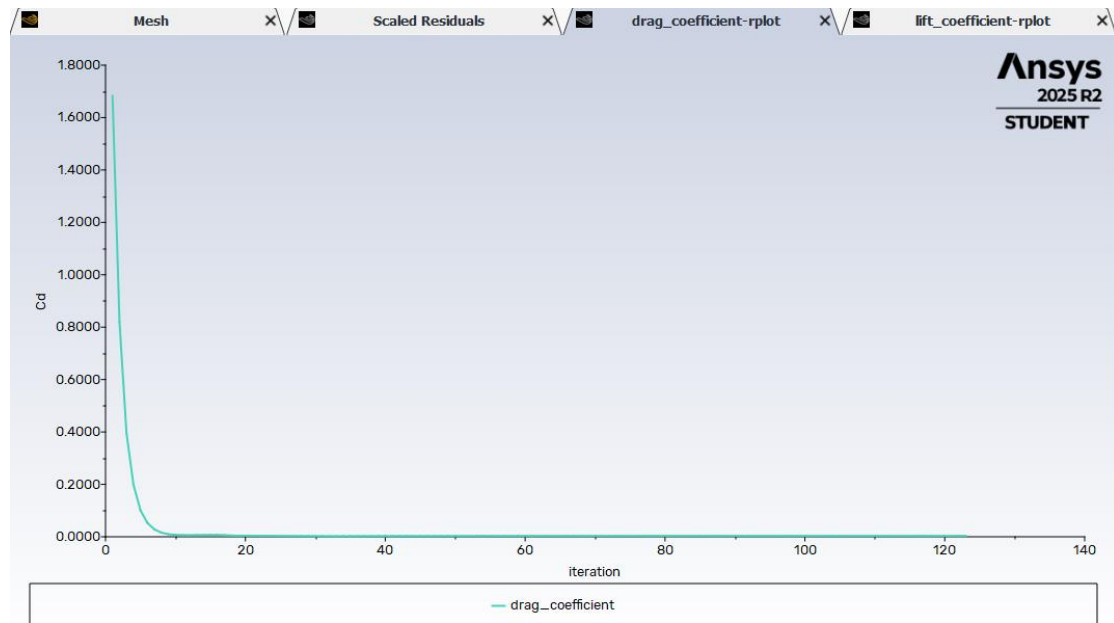
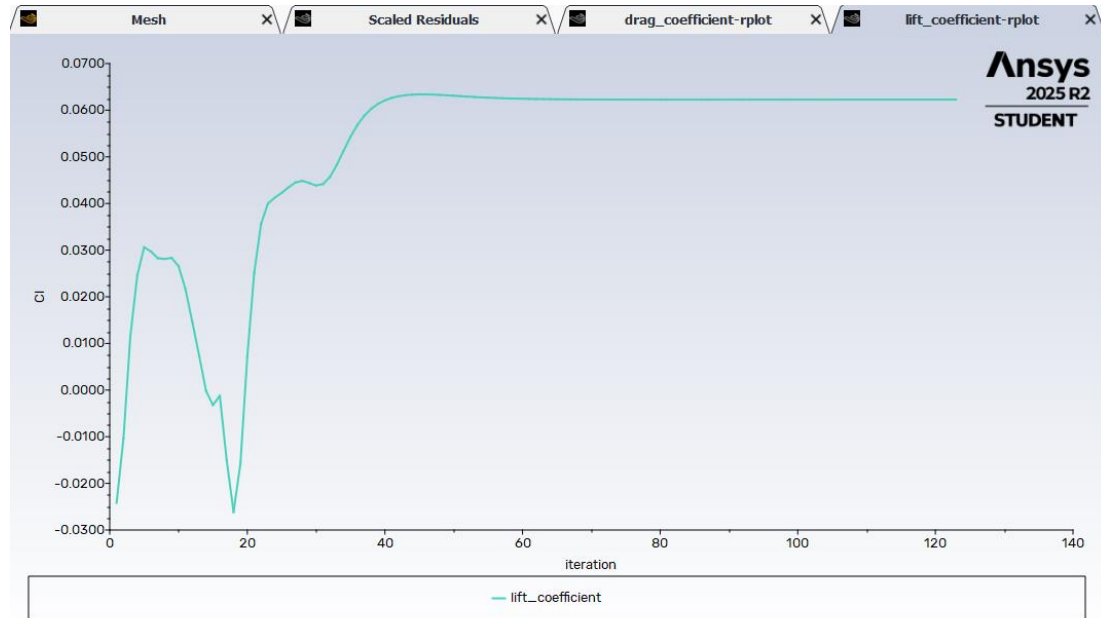
1.2) Final Mesh



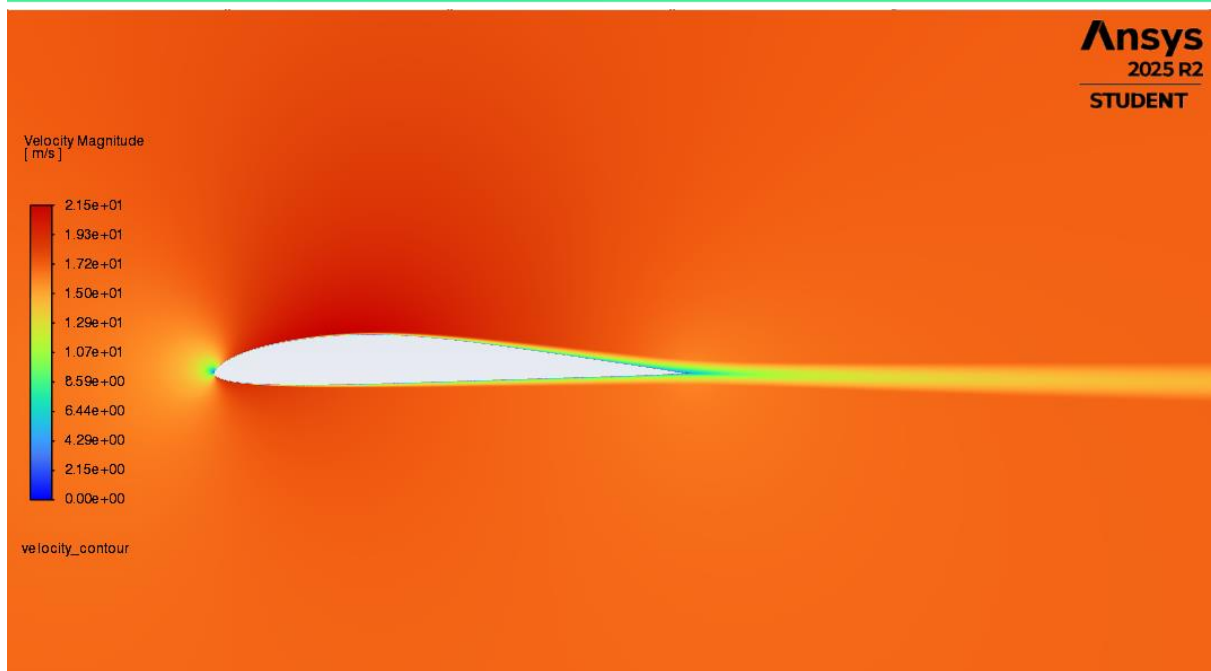
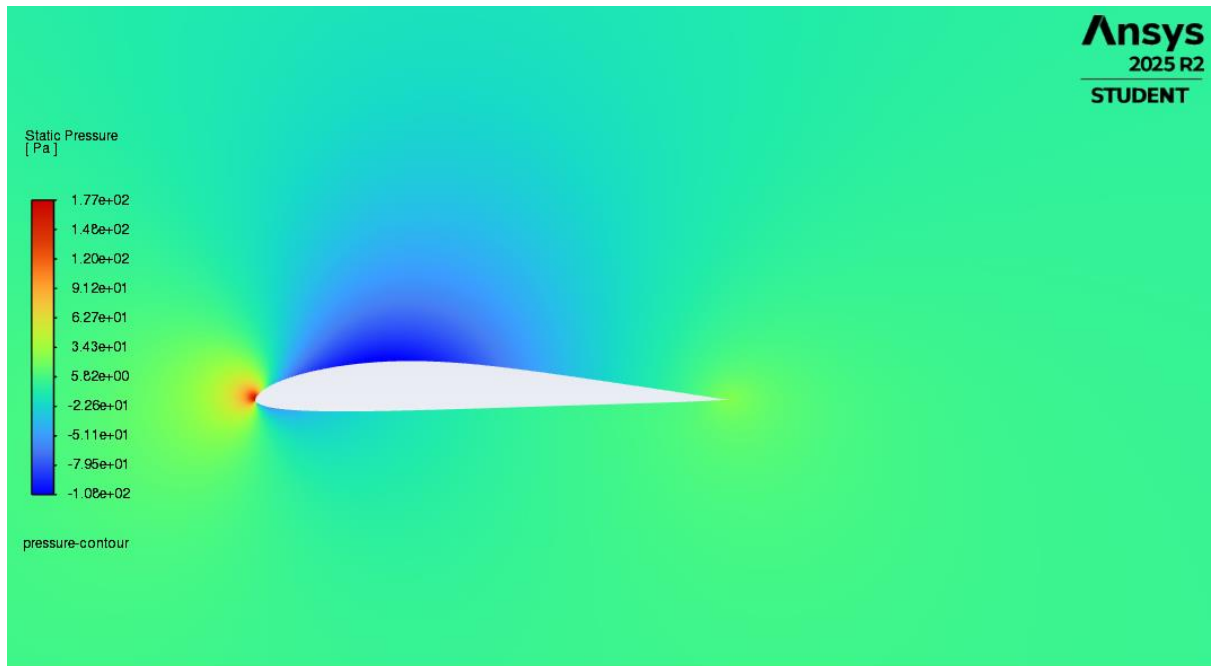
1.3) Named Selections



1.4) Drag/Lift Coefficients



1.5) Pressure/Velocity Contours



2. Laminar Flow Past a Sphere

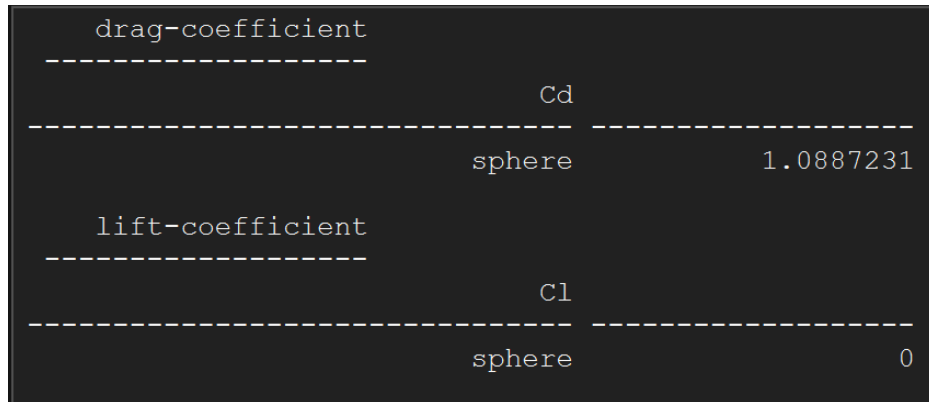
2.1) Axial Velocity Estimate

Axial Velocity at $Re = 100$

$$U_{\infty} = \frac{Re\mu}{\rho D} = \frac{(100)(1.7894 * 10^{-5})}{(1.225)(50 * 10^{-3})} = 0.029 \frac{m}{s}$$

2.6) Reports of Coefficients of Drag & Lift

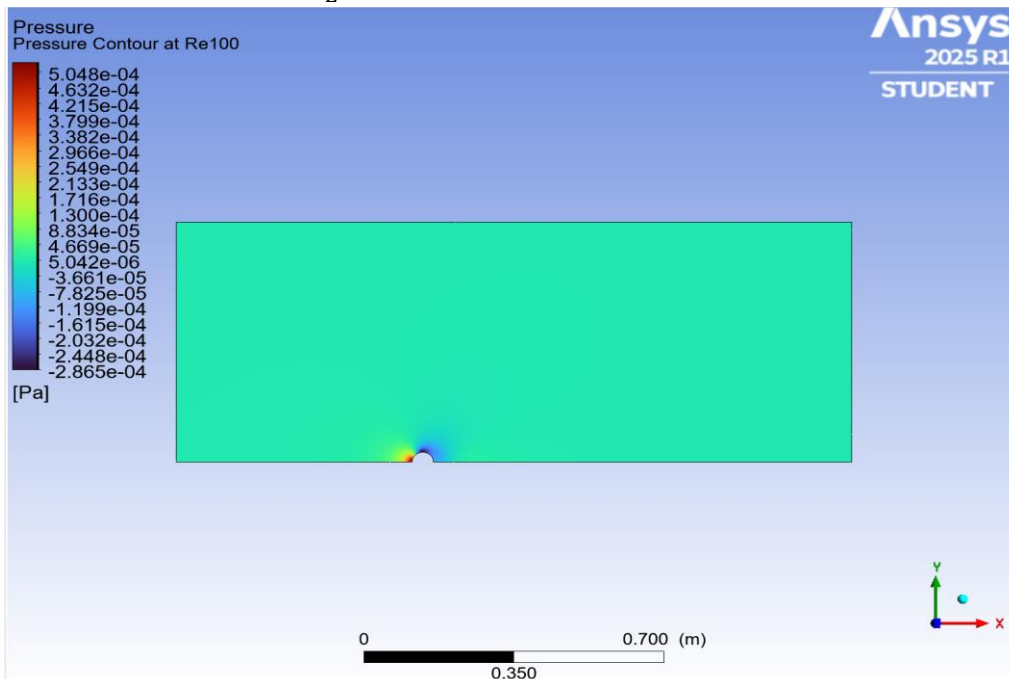
C_d & C_l at $Re = 100$



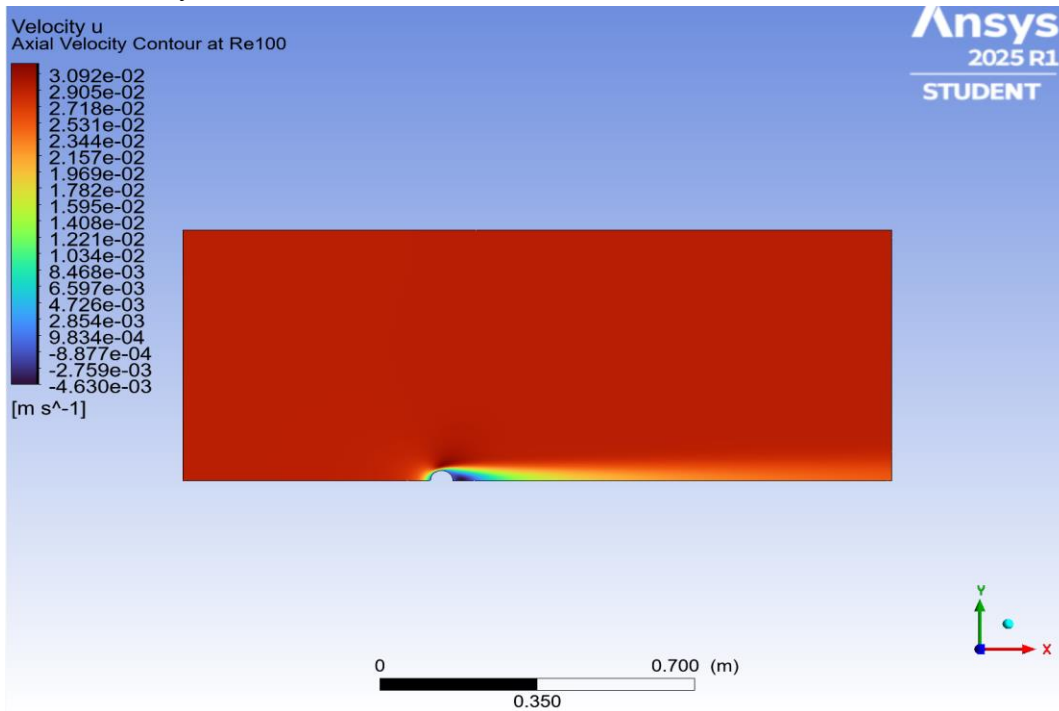
Our answer was very comparable with the experimental drag coefficient, carrying only an 8% error percentage.

2.7) Contours

Pressure Contour Plot at $Re = 100$



Axial Velocity Plot at $Re = 100$



2.8) Repeated Simulations

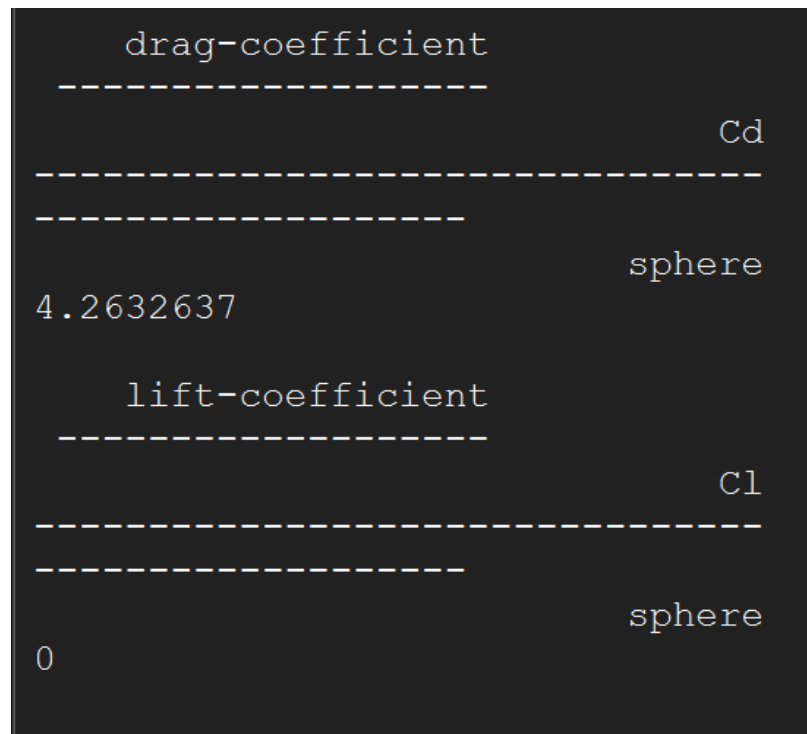
Axial Velocity at $Re = 10$

$$U_{\infty} = \frac{Re\mu}{\rho D} = \frac{(10)(1.7894 \times 10^{-5})}{(1.225)(50 \times 10^{-3})} = 0.0029 \frac{m}{s}$$

Axial Velocity at $Re = 1$

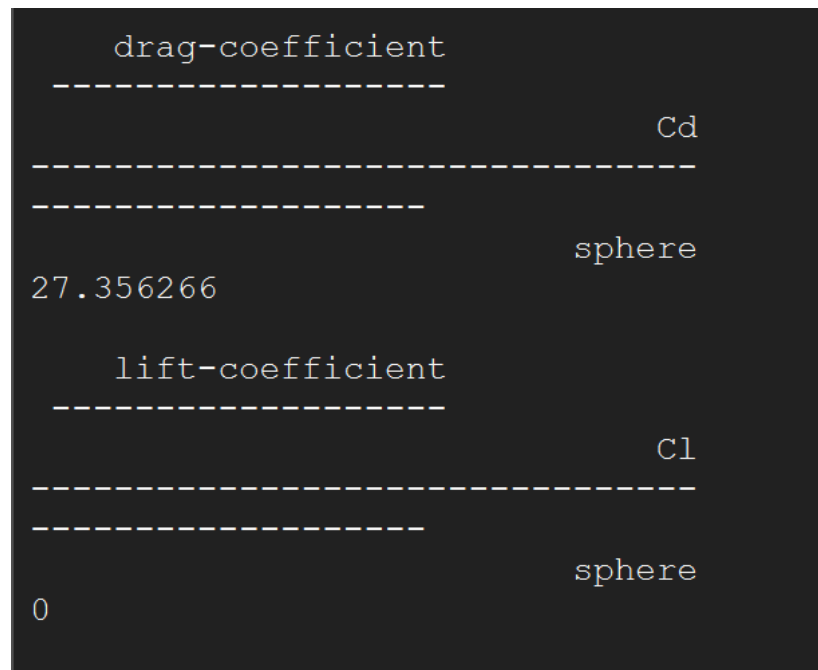
$$U_{\infty} = \frac{Re\mu}{\rho D} = \frac{(100)(1.7894 \times 10^{-5})}{(1.225)(50 \times 10^{-3})} = 0.00029 \frac{m}{s}$$

C_d & C_l at $R_E = 10$



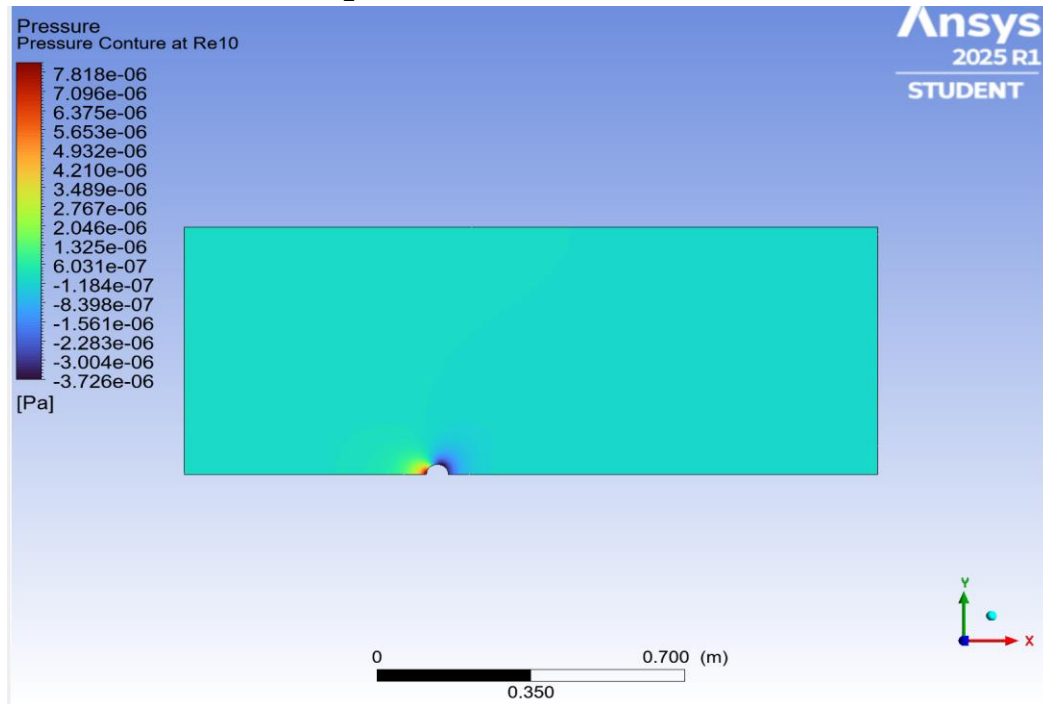
Our answer was much less comparable with the experimental drag coefficient, carrying a 77% error percentage.

C_d & C_l at $R_E = 1$

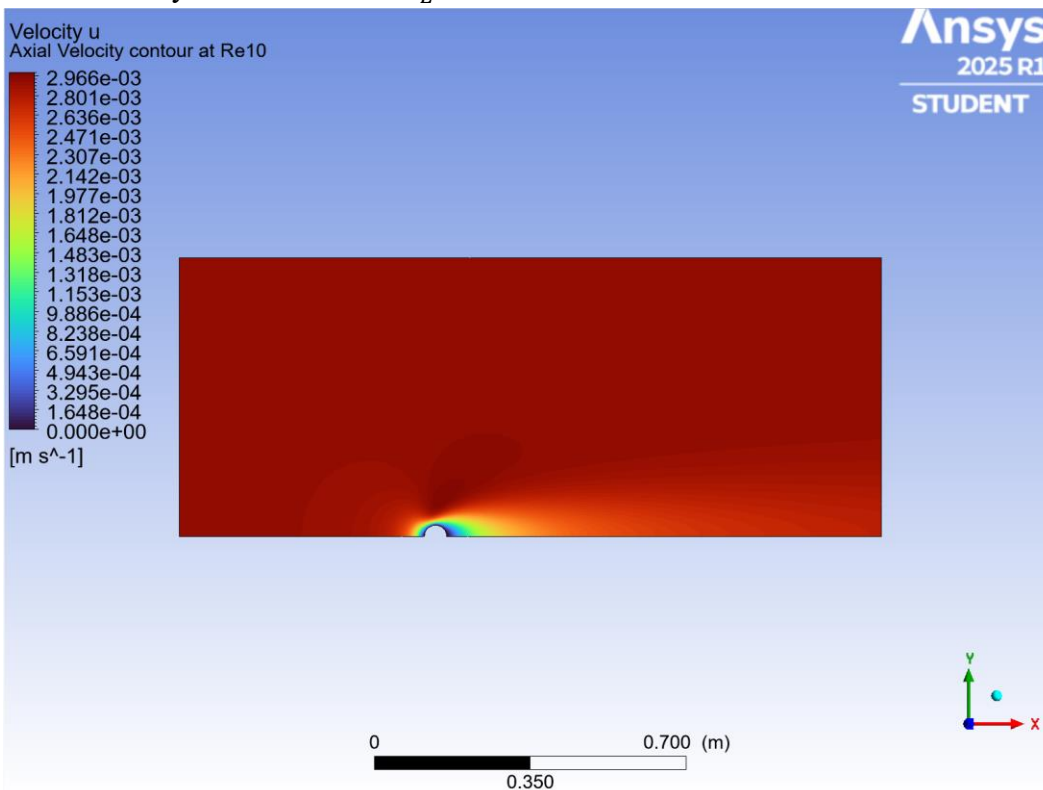


Our answer was comparable with the experimental drag coefficient, carrying only a 14% error percentage.

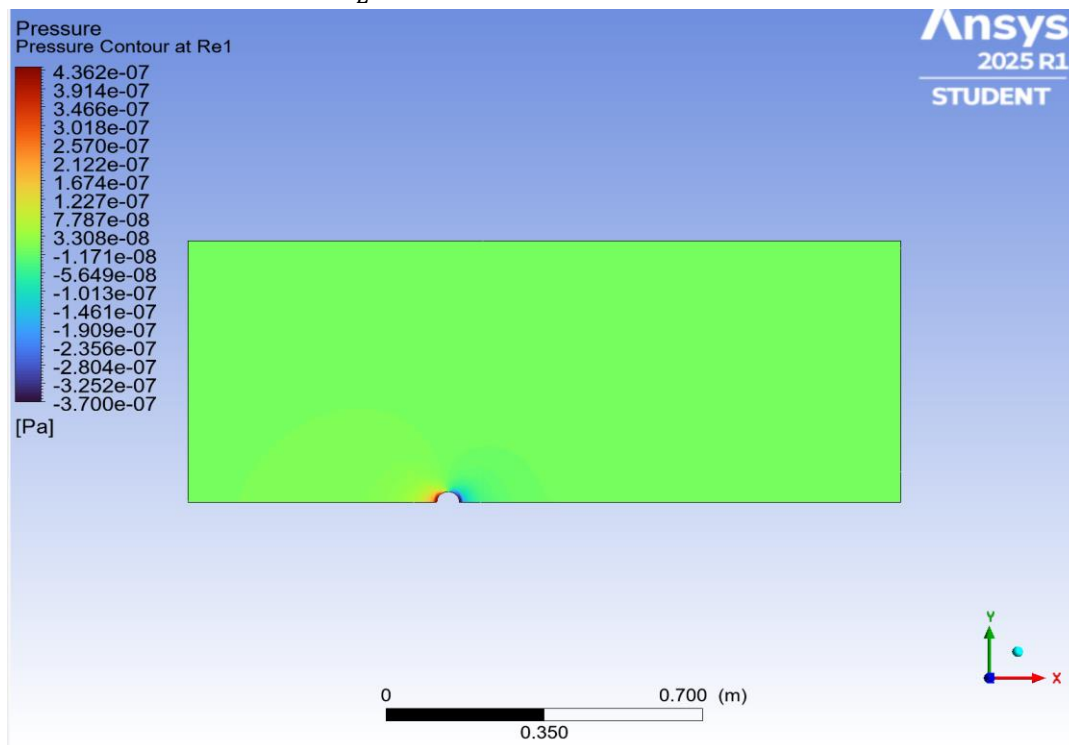
Pressure Contour Plot at $Re = 10$



Axial Velocity Contour Plot at $Re = 10$



Pressure Contour Plot at $R_E = 1$



Axial Velocity Contour Plot at $R_E = 1$

